

Unit I :: Introduction to a Programming Environment

☞ **SEC-01-PHY-91-R-03** ☞

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Syllabus

Unit I: Introduction to a Programming Environment

Software installation and start-up; locating and launching the application, exploring the graphical interface of the software: console, script editor, variable browser; understanding the workspace: current directory, command history, working directory settings; using built-in help and documentation features; basic command execution: arithmetic expressions, variable assignments, managing output display; data types and variables: numeric types (integers, floats), strings, booleans; variable naming conventions; operators and expressions: arithmetic, relational, logical operators; operator precedence and associativity; code structure and readability: using comments, breaking long expressions, line continuation symbols; saving and executing scripts, running code from the script editor or the command window.

1 Setting up the Scilab software

1.1 Installing Scilab

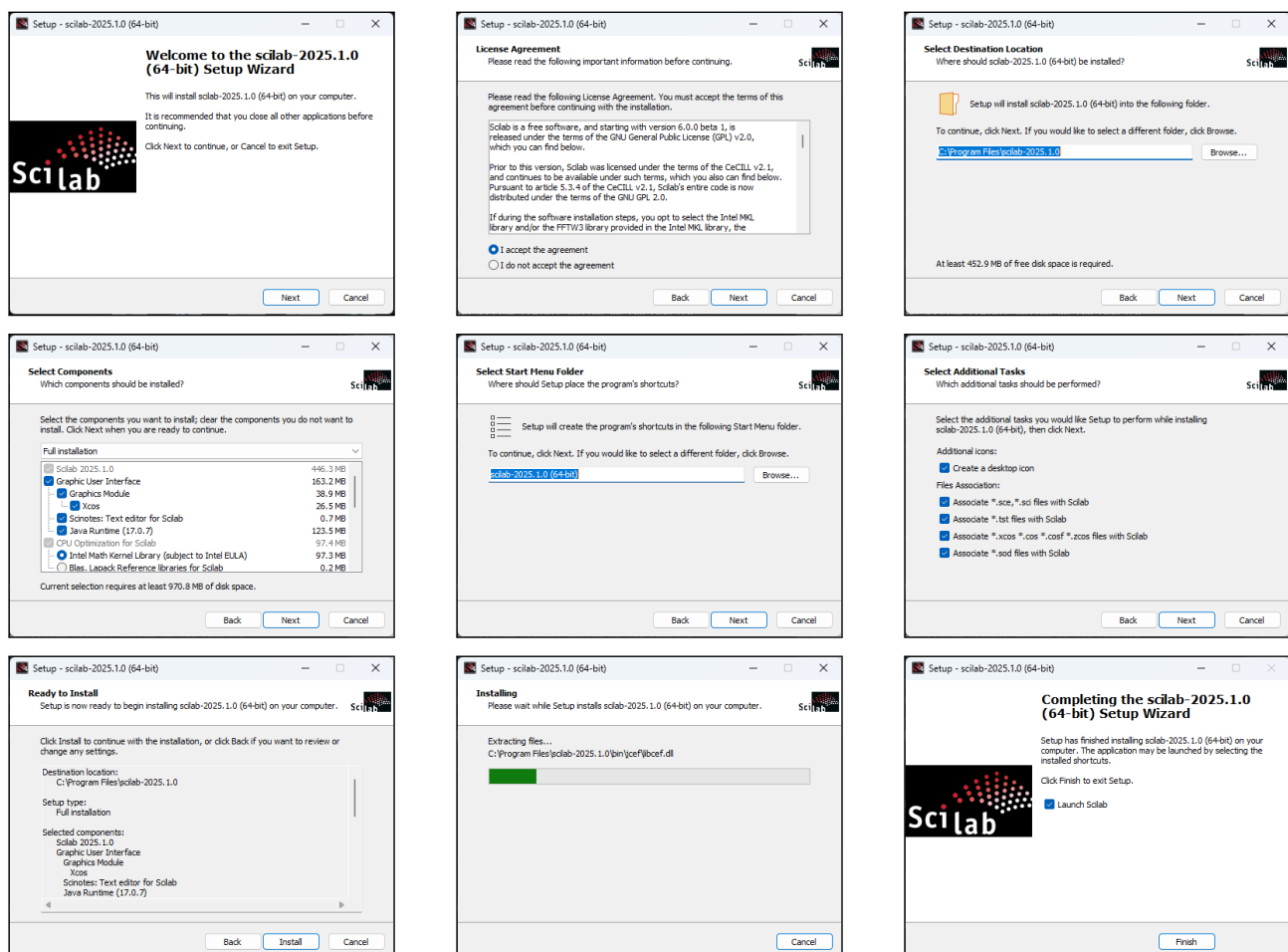
I. Visit the download page of Scilab available at: <https://www.scilab.org/download/>



Figure 1: The webpage for downloading the latest version of Scilab

II. Click on the relevant link to download Scilab for Windows, Linux or MacOS.

III. Navigate to the folder containing the downloaded file and install the software on the PC. Select 'English' as the preferred language the then click on 'OK' to proceed with the rest of the installation process as illustrated below:



1.2 Scilab on Cloud

The *National Virtual Library of India*¹ maintains an online version of Scilab on the Garuda Cloud. This is known as *Scilab on Cloud* and is available at: <https://cloud.scilab.in/>

The online interface of *Scilab on Cloud* is displayed in figure 3.

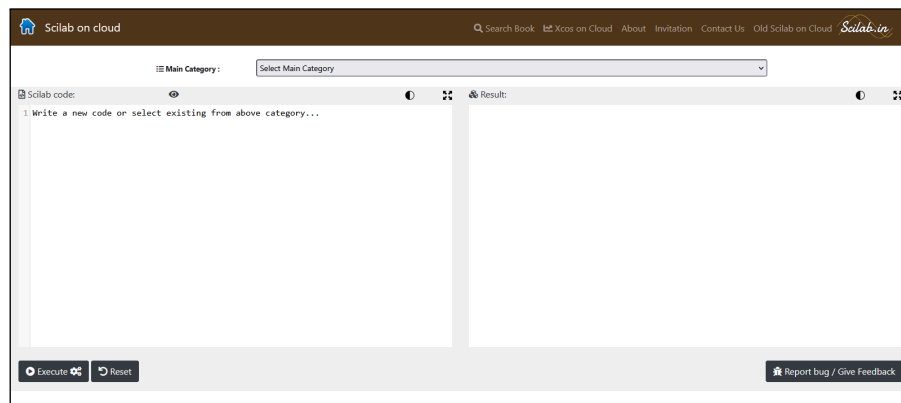


Figure 3: Scilab on Cloud

One can type the Scilab code on the left panel, then click on 'Execute' to view the output on the right panel.

1.3 Launching Scilab

One can find the Scilab icon on the desktop or in the list of installed applications on one's PC. Simply clicking on the icon launches Scilab with its default GUI which is displayed in figure 4 below.

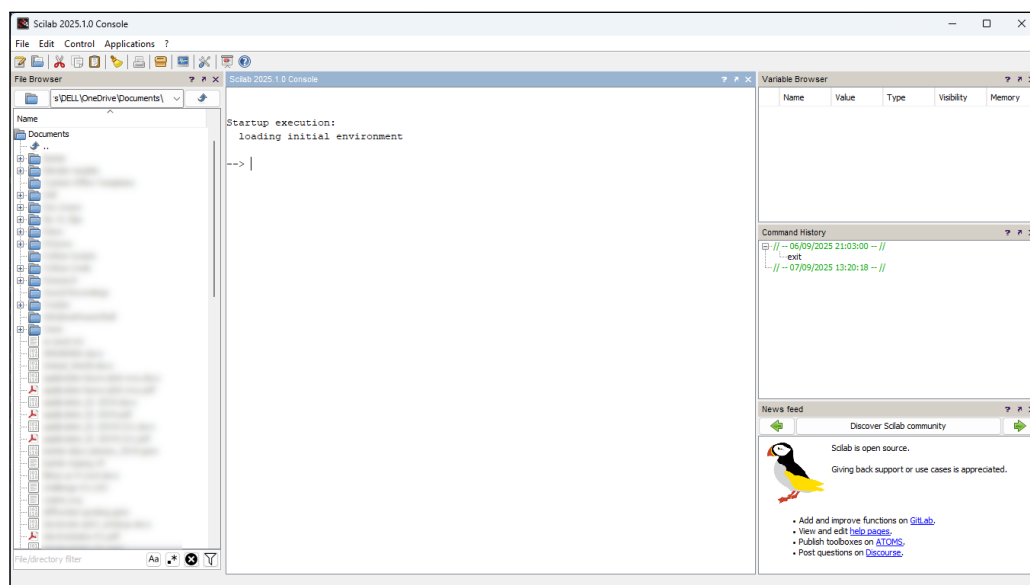


Figure 4: Default graphical user interface (GUI) of Scilab

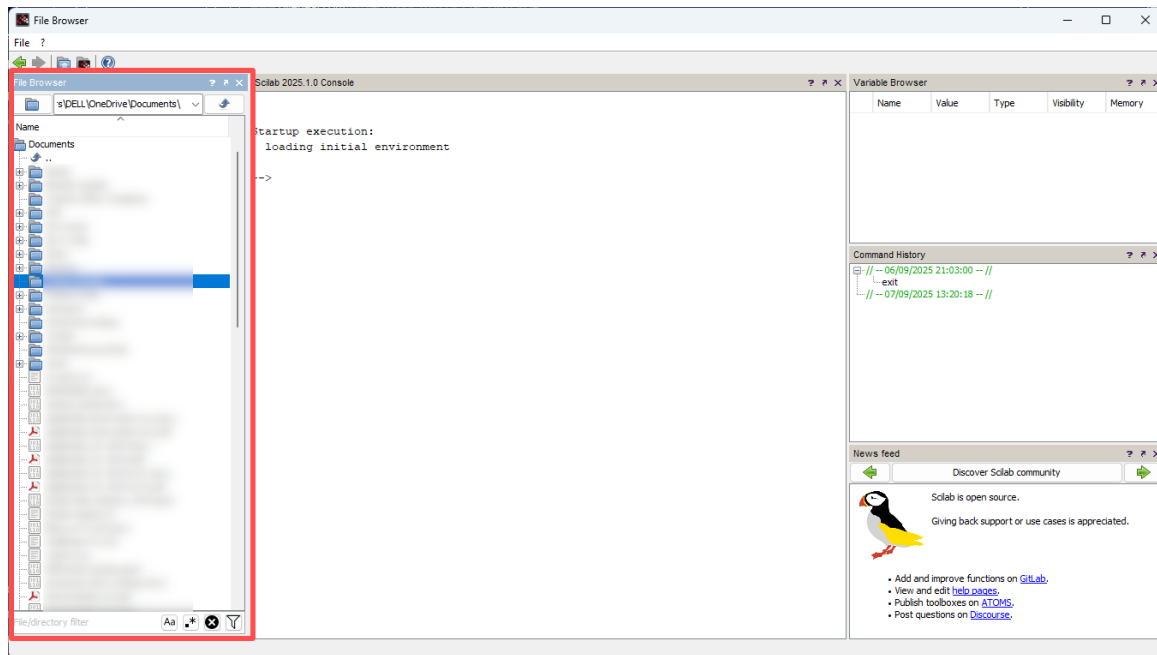
¹<https://www.rrr1f.gov.in/NML/NVLI.aspx>

2 Understanding the GUI of Scilab

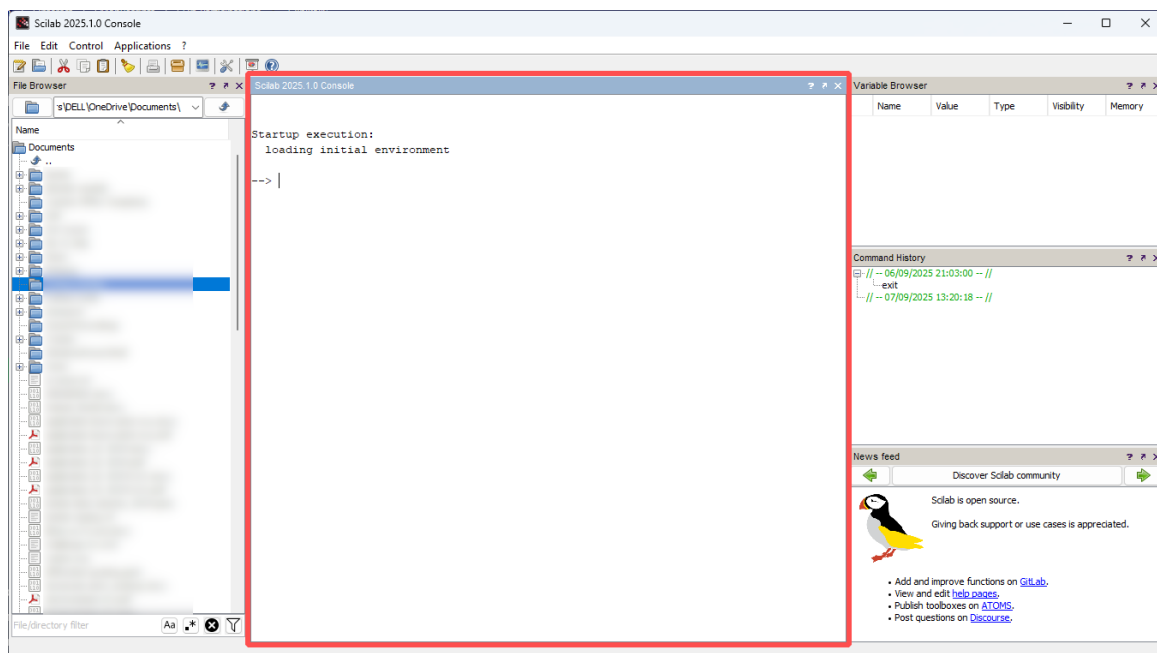
2.1 Panes and layout

The layout of Scilab's GUI is divided into five distinct panes, each of which has its own unique functionality.

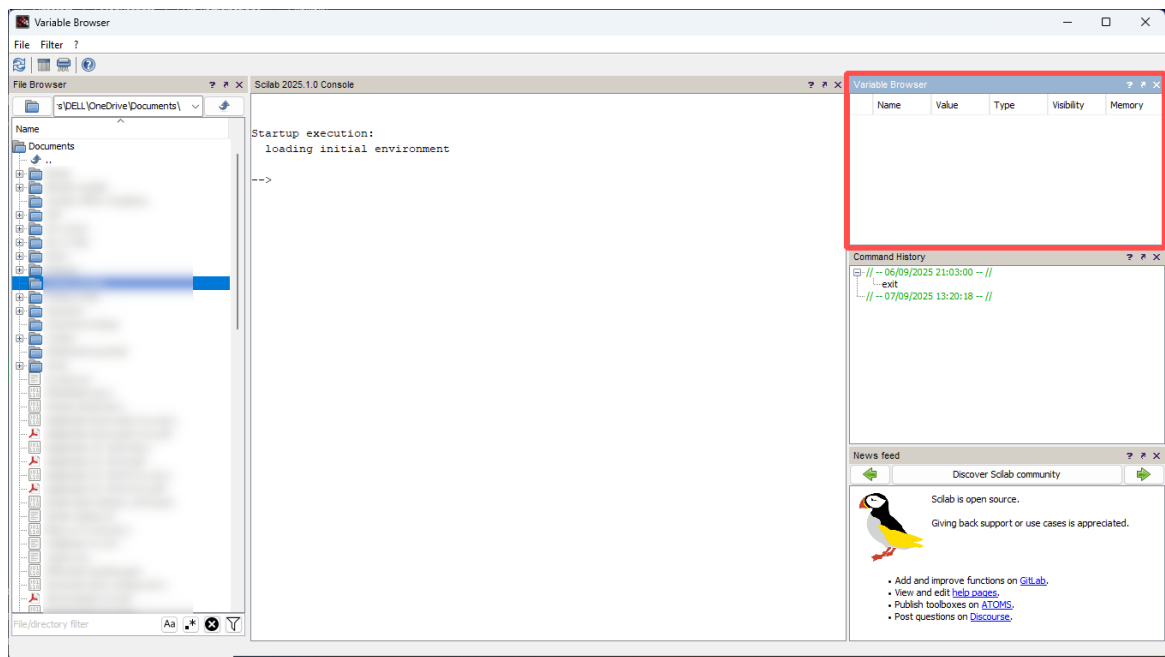
1. *File browser* – allows one to browse through the directory tree to easily navigate to the relevant Scilab script file or folder containing the same.



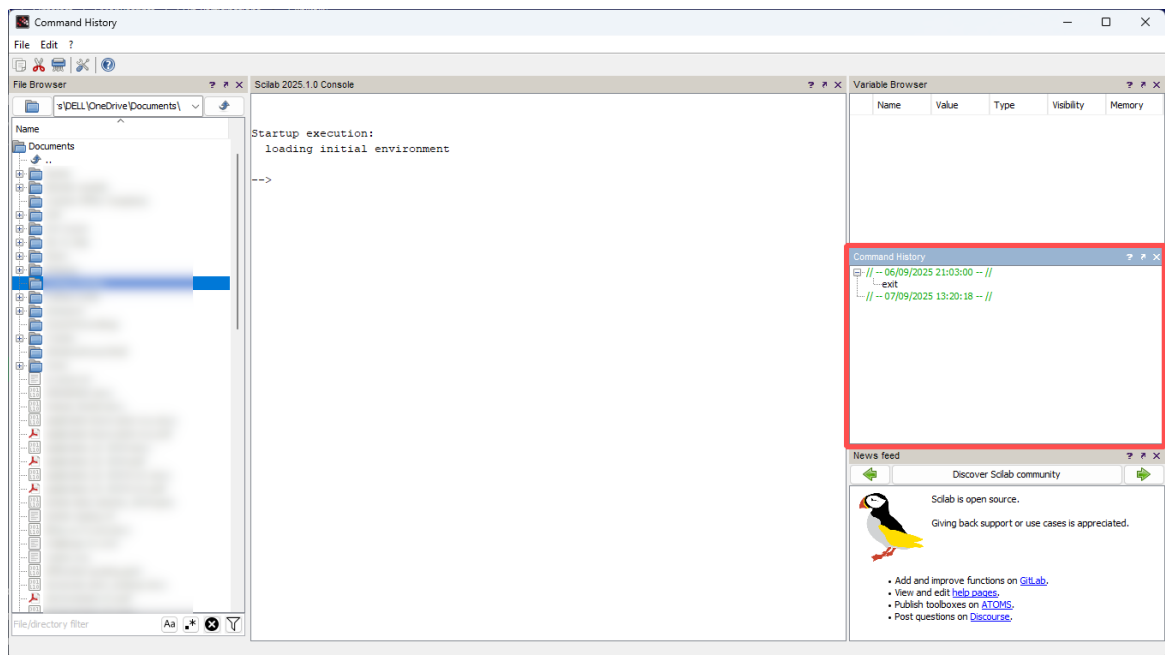
2. *Console window* – allows one to type and execute Scilab commands, as well as to view the output.



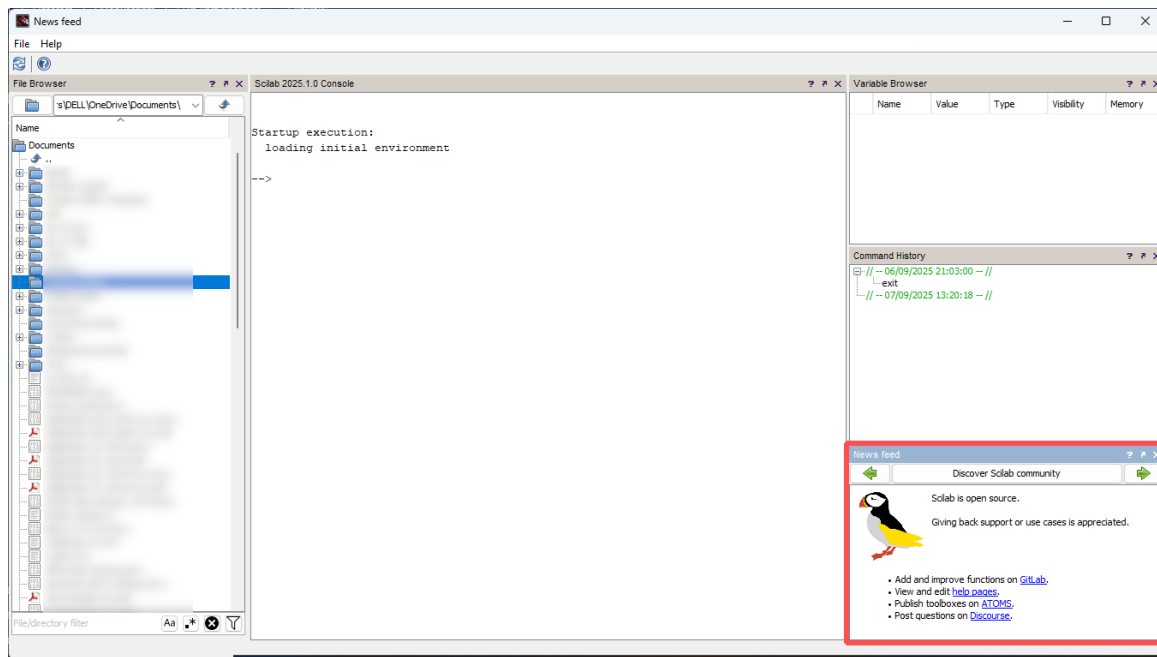
3. *Variable browser* – displays all the variable that have been created by Scilab and are currently active in memory locations.



4. *Command history* – display a chronological order of all the Scilab commands executed by the user, with the most recent command being at the top.



5. *News feed* – provides Scilab news and an interface to interact with the online Scilab community.



2.2 The Workspace

The Scilab workspace consists of the following components:

1. The console window

- Docked on the GUI by default.
- One can perform numerical calculations right from the console and view the output as soon as the computations are complete.
- One can find the current directory that one is working on by using the command `pwd` (which stands for *present working directory*) on the console window.
- The Scilab console presents a full-featured, interactive, command-line REPL (*read-evaluate-print-loop*) shell.

2. The variable browser

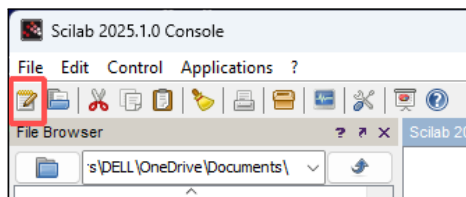
- Docked on the GUI by default.
- One can view the names of the variables, the value they store as well as the memory they are allocated.

3. The command history

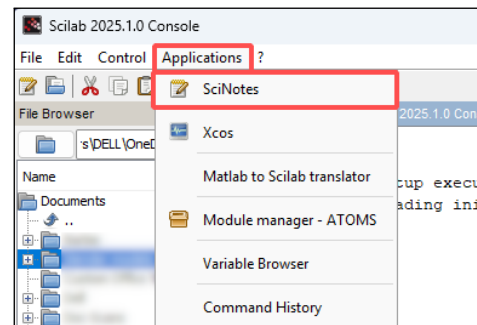
- Docked on the GUI by default.
- One can click on any of the earlier commands from the command history to re-run the same without having to type the command all over again.

4. **The SciNotes editor:** Running multiple lines of command is not possible on the console window. Also, the console window does not allow one to save a program such that it can be executed later. For these reasons, Scilab provides its own editor, known as SciNotes, that allow the user to write and save multi-line programs that can be executed at any point of time on any other PC with Scilab installed.

- Not docked on the GUI by default.
- To open SciNotes, one can either click on its icon on the taskbar (as shown in figure 5a), or one can use the main menu by clicking Applications → SciNotes (as shown in figure 5b).



(a) Opening SciNotes using the taskbar icon



(b) Opening SciNotes using the main menu

Figure 5: Two ways to open SciNotes

5. **The graphic window:** The graphic window is used for displaying any graphical output, such as the plot resulting from a numerical computation or a simulation result.

- Not docked on the GUI by default.
- A graphical window opens up automatically when making any plot.
- A graphical window may be opened by running any of the plot commands in Scilab either from the console window or by including it as a line in a Scilab program written using SciNotes.
- One can view a generic sample plot using a graphical window by simply typing `plot` on the console window, as illustrated in figure 6.

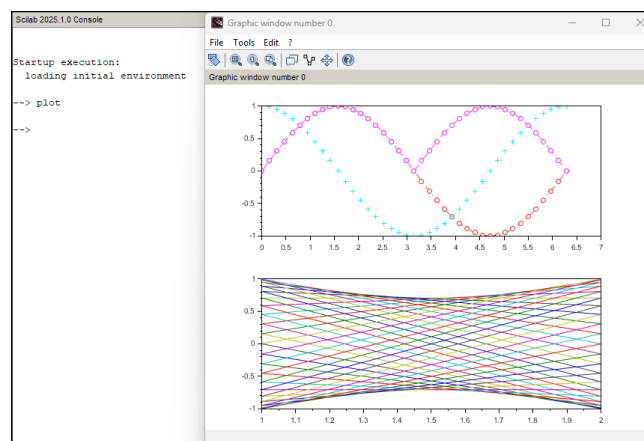


Figure 6: Generic sample plot on a graphical window

2.3 Documentation and help

Scilab provides plenty of resources in the form of documentation that can be easily accessed from the main menu by clicking ?. We are primarily interested in two of the sub-menu items, namely Scilab Help (which can be directly accessed by pressing the F1 key) and Scilab Demonstrations, as illustrated in figure 7.

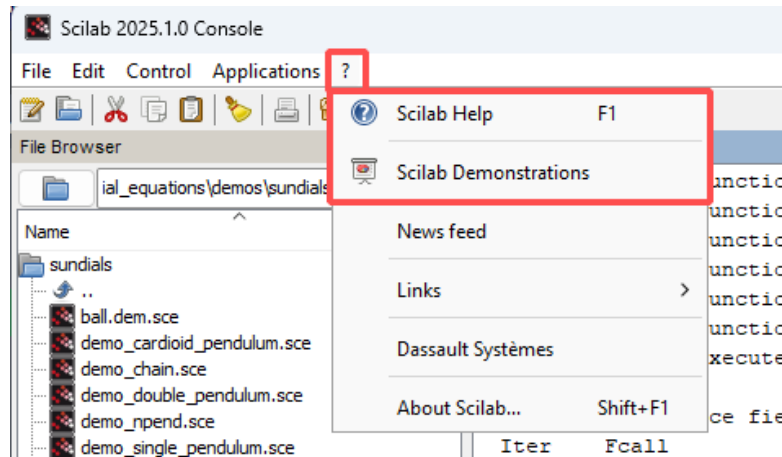


Figure 7: Documentation in Scilab

- **Scilab Help:** The Scilab Help Browser window provides detailed documentation on all the features and functions provided in various toolboxes that come bundled with Scilab.

One can either navigate to the exact topic using the hierarchical directory tree structure, or one may search for the specific topic of interest.

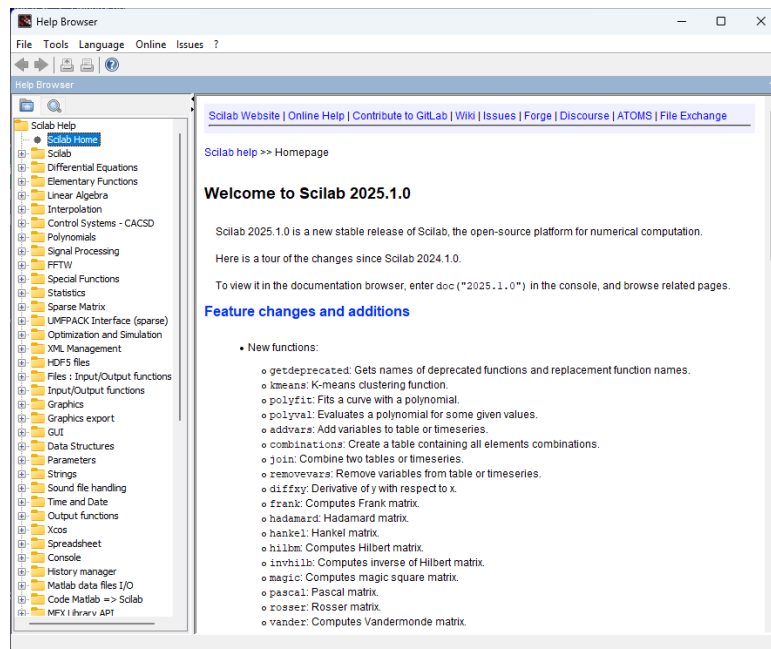


Figure 8: Scilab help browser window

- **Scilab Demonstrations:** Scilab provides a collection of demonstration scripts, which are available on the demonstrations menu. In figure 9, an example is displayed as to how one

can access a particular demonstration, in this case by clicking Simulation → ODE's → Lorenz equation from the demonstrations menu. This would open up a graphic window where the demonstration is performed. One can also view the underlying Scilab code by clicking on -- View Code -- on the main menu of the same graphic window.

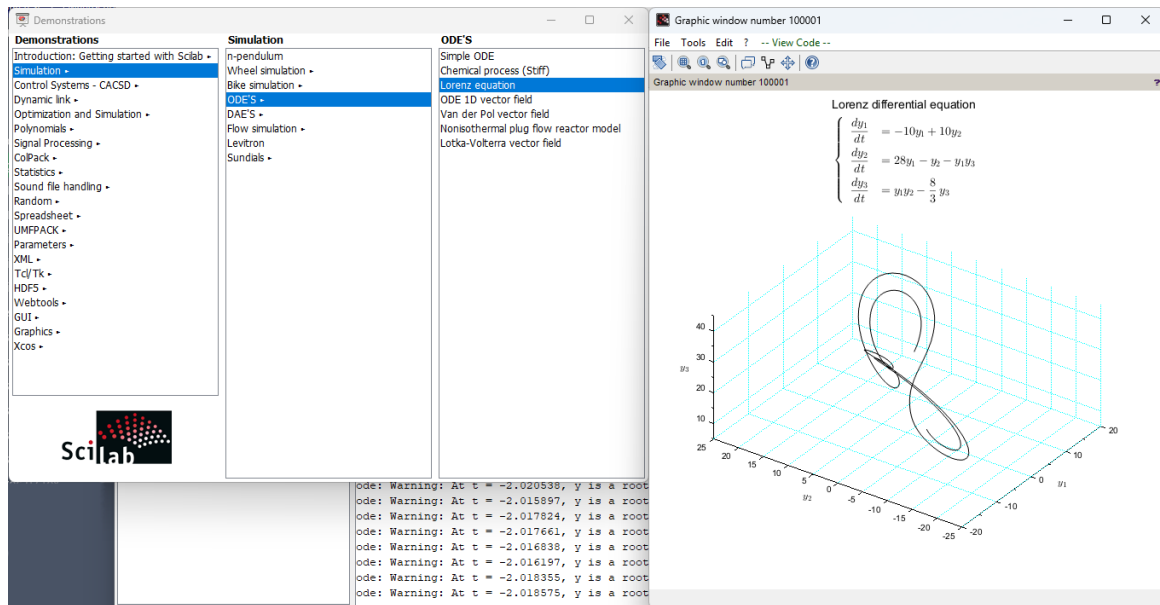


Figure 9: Scilab help browser window

3 Basic command execution

3.1 Arithmetic expressions

Scilab can be used as a simple calculator to perform numerical calculations. Some of such examples are given below.

4 Insert section title

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